Twelve-critical graphs with $\left(\frac{2}{5} - o(1)\right) n^2$ edges

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Abstract

Let $f_k(n)$ denote the maximum number of edges in a k-critical graph on n vertices. We construct a sequence of twelve-critical graphs G_s, with orders tending to infinity, such that $e(G_s)/|V(G_s)|^2 \to 2/5$. This disproves the asymptotic formula $f_{12}(n) \sim 3n^2/8$, and hence the general asymptotic conjecture in Erdős Problem 917 even when the chromatic number is divisible by three. The construction combines K_5-saturated graphs of sublinear maximum degree with a coloring module of Pegden. Its criticality follows from explicit eleven-colorings after the deletion of each type of edge.

1 Introduction

All graphs are finite and simple. A graph is k-critical if it has chromatic number k and every proper subgraph is $(k-1)$-colorable. Write $e(G) = |E(G)|$, and let $f_k(n)$ be the largest number of edges in a k-critical graph on n vertices. One of the questions in Erdős Problem 917 [2] asks whether, for fixed $k \geq 6$,

$$f_k(n) \sim \frac{1}{2}\left(1 - \frac{1}{\lfloor k/3 \rfloor}\right) n^2. \tag{1.1}$$

Known constructions exceed this coefficient when $3 \nmid k$; see the discussion in Luo, Ma, and Yang [3]. For $3 \mid k$, the join of $k/3$ odd cycles of equal order gives the coefficient in (1.1). We show that this coefficient is not optimal for $k = 12$.

Theorem 1. *There is a sequence of twelve-critical graphs G_s such that*

$$|V(G_s)| \longrightarrow \infty, \qquad \frac{e(G_s)}{|V(G_s)|^2} \longrightarrow \frac{2}{5}.$$

In particular, $f_{12}(n) \not\sim 3n^2/8$.

The conclusion concerns the general formula (1.1). It does not resolve the separate question whether $f_6(n) \sim n^2/4$.

The two ingredients are the coloring module $U(S, T)$ introduced by Pegden [4] and the saturated graphs of small maximum degree constructed by Alon, Erdős, Holzman, and Krivelevich [1]. We use five copies of the module. Between their active sets, we take the complement of a graph obtained from a K_5-saturated graph. The maximum-degree condition forces two colors in each active set to be absent from the other four active sets. An eleven-coloring would then require a common additional color in all five sets, contradicting the absence of K_5 in the saturated graph. Saturation supplies the colorings needed after an edge is deleted.

2 A coloring module

We first recall two elementary properties of critical graphs. If F is p-critical and $x \in V(F)$, then F has a p-coloring in which a prescribed color occurs only at x: color $F - x$ with the other $p-1$ colors. Also, the join $F_1 \vee F_2$ of a p-critical graph and a q-critical graph is $(p+q)$-critical. Its chromatic number is $p + q$. Deleting an edge within a factor saves one color. After deleting a joining edge xy, use singleton colors at x and y and identify those two colors, keeping the remaining palettes disjoint. Vertex deletion likewise saves a color in one factor.

Let S be eleven-critical and T be ten-critical. The module $U(S,T)$ consists of disjoint copies of S and T, together with an independent set

$$A = V(S) \times V(T).$$

Each $(x,y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T, and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x.

The following properties are the case $U(11,10)$ of Pegden's module lemma [4, Lemma 2.5]. We include the proof for the graphs used here; it does not require the factors to be triangle-free.

Lemma 2. *Fix a palette of eleven colors.*

1. *In every proper coloring of $U(S,T)$ from this palette, the active set uses at least three colors.*

2. *Given $a_0 = (x_0,y_0) \in A$ and distinct colors α, β, γ, there is a proper coloring in which the active set uses only α, β, γ, and γ occurs on the active set only at a_0.*

3. *Given an edge e of $U(S,T)$ and distinct colors α, β, there is a proper coloring of $U(S,T) - e$ in which the active set uses only α, β.*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all eleven colors, it contains a vertex colored α and a vertex colored β. Every active vertex in the first row must have color β, so no vertex of T can have color β. The second row similarly excludes α from T. This leaves only nine colors for T, a contradiction.

For (2), color S so that α occurs only at x_0. Color T with the ten colors other than α, so that β occurs only at y_0. Extend these colorings by

$$\operatorname{col}(x,y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0,\ y \neq y_0, \\ \gamma, & (x,y) = (x_0,y_0). \end{cases} \tag{2.1}$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the ten colors other than α, and color all of A with α. If $e \in E(T)$, color $T - e$ with the nine colors other than α, β. Choose $x_0 \in V(S)$ and color S with α only at x_0. Color the row indexed by x_0 with β and all other rows with α.

Suppose instead that e joins an active vertex (x_0,y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0,y_0) to x_0, use (2.1) with the color of (x_0,y_0) changed to α. If it joins (x_0,y_0) to y_0, change that color to β. In either case the only edge that would become monochromatic is the deleted edge. □

3 From saturated graphs to twelve-critical graphs

A graph is K_5-saturated if it contains no K_5, but adding any missing edge creates a K_5. Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a triangle.

Proposition 3. *Let H be a K_5-saturated graph on $v \geq 5$ vertices, with maximum degree $d < v-1$. Let $h \geq 11$ be odd and suppose*

$$v > 40hd. \tag{3.1}$$

Put $a = 2hv$. There is a twelve-critical graph G with

$$|V(G)| = 5(a+h+2v) \tag{3.2}$$

and

$$e(G) \geq 10a^2\left(1-\frac{d}{v}\right). \tag{3.3}$$

More precisely, if $m = e(H)$, then

$$e(G) = 10(a^2 - 8h^2m) + 5(2a+9h+16v-79). \tag{3.4}$$

Proof. Construction. Set $\ell = 2v$ and take

$$S = K_8 \vee C_{h-8}, \qquad T = K_7 \vee C_{\ell-7}.$$

Both cycles are odd and have length at least three. Hence S and T are eleven-critical and ten-critical, respectively, with h and ℓ vertices.

Take five disjoint copies $U_1, \ldots, U_5$ of $U(S,T)$, with active sets $A_1, \ldots, A_5$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0,1\}$ by any bijection. An active vertex then has the form $(x,(z,\varepsilon))$; give it label z. Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a five-partite graph Z on $A_1 \cup \cdots \cup A_5$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \quad \Longleftrightarrow \quad \pi(u)\pi(w) \in E(H). \tag{3.5}$$

The graph G retains all the edges of the five modules and, between different active sets, takes the complement of Z. There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G.

We record three properties of Z. First, Z is K_5-free: the labeling map sends every clique injectively to a clique of H. Moreover, adding any missing edge between different parts of Z creates a K_5 with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a triangle in $N_H(z) \cap N_H(z')$. Choose vertices with these three labels in the other three parts. Together with u, w, they form the required clique in $Z + uw$. If the labels coincide, say both equal z, choose a vertex z' not adjacent to z; this is possible because $d < v-1$. The same common-neighborhood triangle is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_4 across any prescribed four parts. Indeed, H has a missing edge; one endpoint together with a triangle in the common neighborhood gives a K_4 in H. Its four labels can be placed in any four parts of Z.

Third, writing $D = \Delta(Z)$, condition (3.1) gives

$$D = 4bd = 8hd, \qquad \ell > 10D, \tag{3.6}$$

The chromatic lower bound. Suppose that G has an eleven-coloring. In each A_i, at least two color classes have size greater than D. Otherwise, choose the unique color with more than D vertices, if there is one, and call it α; if there is none, choose any color. The other ten colors together occupy at most $10D < \ell$ vertices of A_i. Every row, having ℓ vertices, therefore contains a vertex of color α. Each vertex of the copy of S is adjacent to its entire row, so this copy of S would avoid α, contradicting $\chi(S) = 11$.

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Choose two such colors in each part. These are ten distinct colors, each absent from the other four active sets. By Lemma 2(1), every active set needs a third color. The one remaining color must consequently occur in all five active sets. Choosing a vertex of this color in each part gives a K_5 in Z, a contradiction. Thus $\chi(G) \geq 12$.

Colorings after edge deletion. Use an eleven-color palette

$$\alpha_1, \beta_1, \ldots, \alpha_5, \beta_5, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G. By the first property of Z, there is a K_5 in $Z + uw$ containing uw, with vertices $a_i \in A_i$. Apply Lemma 2(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i. The only color shared by different active sets is γ. The five vertices of this color are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained an eleven-coloring of $G - e$.

Now let e lie within U_i. Apply Lemma 2(3) to color $U_i - e$, using only α_i, β_i on A_i. Choose a K_4 in Z across the other four parts. In each of those four modules, apply Lemma 2(2) with the selected vertex as the only active vertex of color γ. Once again this is a proper eleven-coloring of $G - e$.

These cases include every edge of G. Recoloring one endpoint of a deleted edge with a twelfth color gives $\chi(G) \leq 12$. Thus $\chi(G) = 12$, and deletion of every edge lowers the chromatic number. The graph has no isolated vertices, so a subgraph missing a vertex is also contained in some $G - e$. Consequently G is twelve-critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each. Thus the number of edges of G between active sets is

$$10(a^2 - 2b^2m) = 10(a^2 - 8h^2m).$$

The two structural graphs have

$$e(S) = 9h - 44, \qquad e(T) = 16v - 35,$$

and each module has $2a$ further edges incident with its active set. This proves (3.4). Finally, $2m \leq vd$ and $a = bv$ give (3.3) from the edges between active sets alone. □

4 Choice of the saturated graphs

We use the following instance of the construction in [1, Section 4, Example 2].

Lemma 4 (Alon–Erdős–Holzman–Krivelevich)**.** *For every prime power $Q \geq 4$, there is a K_5-saturated graph H_Q with*

$$|V(H_Q)| = 13Q^2 + 12Q, \qquad \Delta(H_Q) = 22Q - 3. \tag{4.1}$$

To obtain this graph from Example 2 of [1], set $k = 5$ and give each of the Q^2 additional independent sets one vertex. The core has $12Q(Q+1)$ vertices. Each core vertex has $3(3Q-1)$ neighbors in its level, $12Q$ in the other levels, and Q among the additional vertices. Its degree is therefore $22Q - 3$. Each additional vertex has degree $12(Q+1)$, which is smaller for $Q \geq 4$. This gives (4.1); saturation is part of the cited construction.

Proof of Theorem 1. For each integer $s \geq 4$, set

$$h = 3^s, \qquad Q = h^2,$$

and take $H = H_Q$ from Lemma 4. Write $v = |V(H)|$ and $d = \Delta(H)$. Then

$$v = 13h^4 + 12h^2, \qquad d = 22h^2 - 3 < v - 1.$$

The integer h is odd and at least 81. Moreover,

$$40hd < 880h^3 < 13h^4 \leq v,$$

since $13h \geq 13 \cdot 81 > 880$. Proposition 3 therefore gives a twelve-critical graph G_s. Put $a = 2hv$ and $N_s = |V(G_s)|$. We have

$$\frac{N_s}{5a} = 1 + \frac{1}{h} + \frac{1}{2v} \longrightarrow 1, \qquad \frac{d}{v} \longrightarrow 0. \tag{4.2}$$

There are at most $10a^2$ edges between active sets, and the five modules contain $5(2a+9h+16v-79) = O(a)$ edges. Together with (3.3), this gives

$$10a^2\left(1 - \frac{d}{v}\right) \leq e(G_s) \leq 10a^2 + O(a).$$

Dividing by N_s^2 and using (4.2) yields

$$\frac{e(G_s)}{N_s^2} \longrightarrow \frac{10}{25} = \frac{2}{5}.$$

Finally, $N_s \to \infty$ and $f_{12}(N_s) \geq e(G_s)$. Since $2/5 > 3/8$, this subsequence rules out $f_{12}(n) \sim 3n^2/8$. □

AI tool disclosure

OpenAI Codex (GPT-6) was used substantially in the development of this work, including the composition of saturated graphs with the coloring module, the proof of criticality, literature retrieval, and preparation and checking of the manuscript. The author selected the problem and directed the research. An exact-arithmetic script was also used to check the parameter counts; the proof is independent of that computation. The author is responsible for the mathematical content. This draft is circulated for review, and the AI-assisted checks are not presented as independent human verification.

References

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